1 Networks

1.1 Toy Network (two-corridor)

Topology

The *quickstart* toy network is a two-corridor graph with two origin TAZs (91, 92) and two destination TAZs (93, 94), connected through six internal junctions (1–6) and four intermediate single-degree auxiliary nodes (911–914). The cross-links `L15` (6 → 3) and `L16` (3 → 6) couple the two corridors. Figure 1 shows all raw SUMO edge IDs; Figure 2 shows the same network after edge grouping, which yields **6 grouped flow variables**, **4 OD pairs**, and **4 routes** in total.

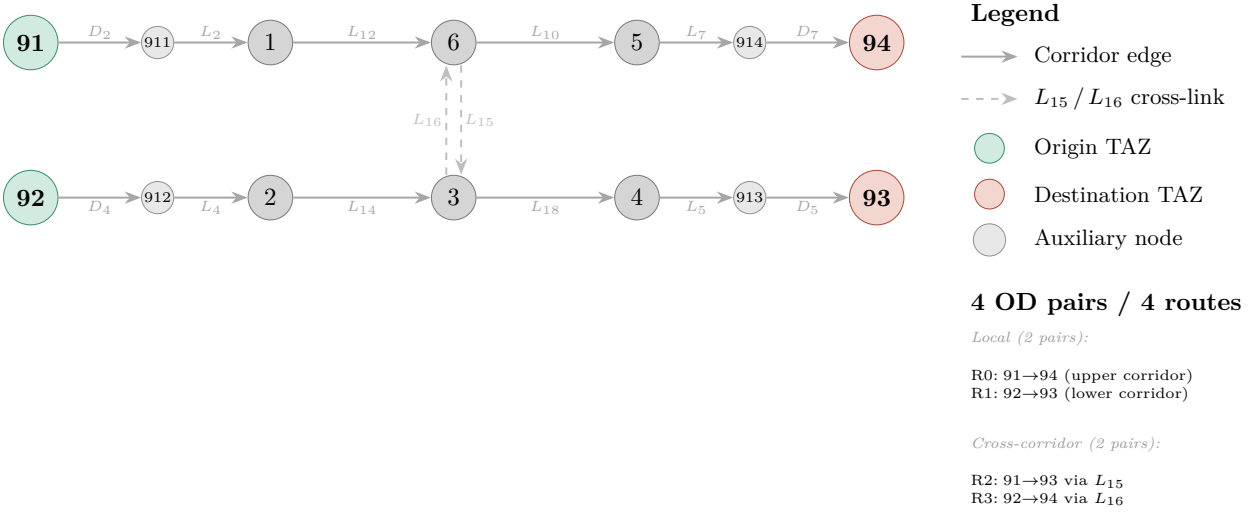


Figure 1: Toy network with all raw SUMO edge IDs. Auxiliary nodes 911–914 are single-degree merge points that carry identical flow to their adjacent edge and are therefore absorbed into grouped flow variables (see Figure 2).

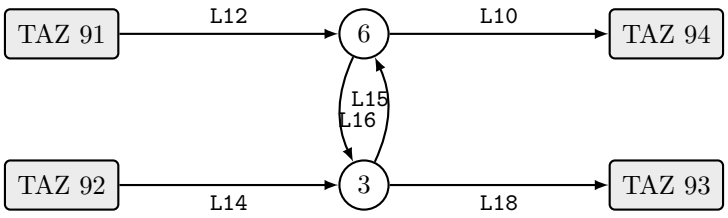


Figure 2: Toy network after edge grouping. Each group collapses a chain of raw edges into a single flow variable: `L12` = {D2, L2, L12}, `L10` = {D7, L7, L10}, `L14` = {D4, L4, L14}, `L18` = {D5, L5, L18}. Intermediate junctions (1, 2, 4, 5) and auxiliary nodes (911–914) are absorbed and no longer appear.

1.2 Scaled Network (`quickstart_grid_3`)

Topology

The `quickstart_grid_3` network is constructed by replicating the two-corridor base module three times (copies 0, 1, 2) and connecting adjacent copies at their left and right interior junctions via bidirectional *green* and *purple* cross-streets (Figure 3). This yields 6 origin TAZs, 6 destination TAZs, **36 grouped flow variables**, **16 OD pairs**, and **20 routes** in total.

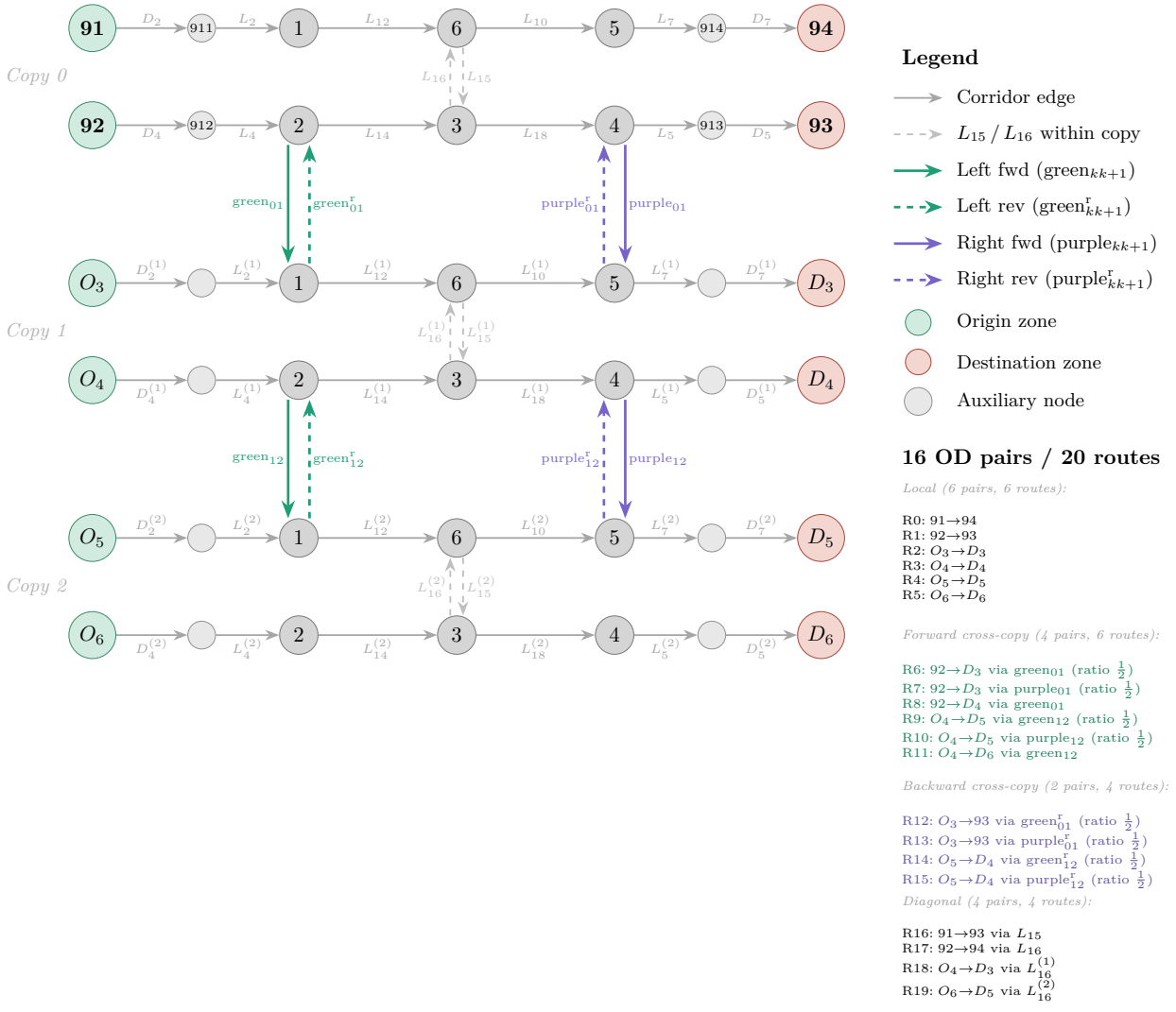


Figure 3: Three-corridor synthetic network (`quickstart_grid_3`). Green and purple arrows denote cross-corridor connectors; dashed arrows within each corridor are the L_{15} / L_{16} diagonal links.

1.3 Grid Network (`grid_2d_3x3`)

Topology

The `grid_2d_3x3` network is a 3×3 grid with nine junctions arranged in three rows (A, B, C) and three columns (0, 1, 2), connected by edges along every adjacent pair. The four corner junctions (A0, A2, C0, C2) each connect to a dedicated TAZ that generates and absorbs traffic; the five remaining junctions (A1, B0, B1, B2, C1) are purely internal. The central junction B1 acts as a bottleneck through which all cross-diagonal demand paths must pass. Each of the **24 directed edges** is its own flow variable (no grouping required). The four corner TAZs define **12 OD pairs**, each served by 4 path routes at an equal split ratio of 0.25, giving **48 routes** in total. Figure 4 shows the full topology.

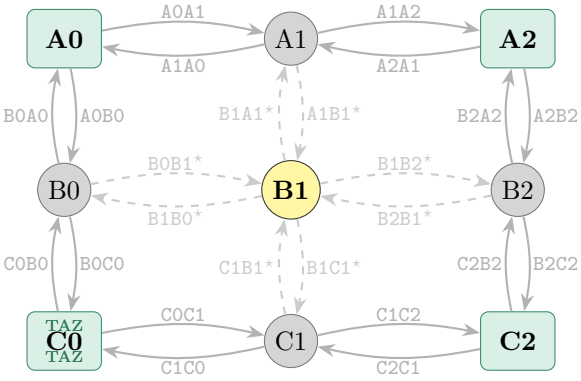


Figure 4: Grid network (`grid_2d_3x3`).